Rockchip Linux Memory Allocator Developer Guide

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Preface

Overview

This document introduces technologies related to the memory allocator on the Rockchip Linux 5.10 platform.

Linux 5.10 introduces the DMA-HEAP memory allocator, with code located in the `drivers/dma-buf/heaps/` directory. It is dedicated to allocating DMABUF-type memory and returns the dmabuf file descriptor (fd) to user space via ioctl.

Product Version

Chip Models	Kernel Version
RK3588, RV1106	Linux-5.10

Intended Audience

This document (this guide) is primarily intended for the following engineers:

- Technical Support Engineers
- Software Development Engineers

Revision History

Version	Author	Date	Description
V1.0.0	Xu Jianqun	2022-06-25	Initial version

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1. Allocator 1: DMA-HEAP

The allocator driver code is located in the `drivers/dma-buf/heaps/` directory.

```
drivers/dma-buf/heaps/
├── cma_heap.c  Contiguous Memory Allocator driver
├── deferred-free-helper.c  Driver for deferred memory release
├── deferred-free-helper.h
├── Kconfig
├── Makefile
├── page_pool.c  Memory page cache Driver
├── page_pool.h
└── system_heap.c  Driver for the non-contiguous physical address and contiguous virtual
address memory allocator
```

A heap driver represents an allocator for a specific memory type and supports modular compilation. Linux provides two typical memory allocator drivers: system-heap and cma-heap.

1.1 system-heap

The system-heap supports CPU cache by default. Memory allocated by this allocator is cached for CPU access, enabling fast access speeds; however, data consistency between the device and the CPU must be maintained via software operations. This driver extends to an allocator that does not support CPU cache, known as system-uncached-heap.

The system-heap supports defer-free by default, meaning memory release is delayed. When the memory page cache is non-empty, memory allocation is fast. The memory page cache relies on the mm shrinker for appropriate reclamation.

The system-heap adds an allocator to the dma-heap via add_dma_heap(), obtains a unique index, and creates the allocator device under `/dev/dma_heap/`, such as `/dev/dma_heap/system-heap` and `/dev/dma_heap/system-uncached-heap`.

1.2 cma-heap

The cma-heap supports CPU cache by default. Memory allocated by this allocator is cached for CPU access, enabling fast access speeds; however, data consistency between the device and the CPU must be maintained via software operations.

The cma-heap allocates memory pages from cma_alloc. The CMA memory region is determined by the CMA driver; for details, refer to the `Rockchip_Developer_Guide_Linux_CMA_CN` document. This allocator provides physically contiguous memory, suitable for direct access by peripherals without an IOMMU.

2. Allocator 2: DRM-GEM

The allocator Driver code is located in the `drivers/gpu/drm/rockchip/` directory.

3. Allocator 3: VB2-MEMOPS

The allocator Driver code is located in the `drivers/media/common/videobuf2/` directory.

4. RV1106-Specific Allocator: RK-DMA-HEAP

RK-DMA-HEAP is a DMABUF memory allocator specific to the RV1106, based on DMA-HEAP.

The main differences between RK-DMA-HEAP and DMA-HEAP are:

	Linux API	RETURN
Linux DMABUF Allocation	rk_dma_heap_buffer_alloc	struct dma_buf *
Linux PAGE Array Allocation	rk_dma_heap_alloc_contig_pages	struct page *
Debug List Support		

> DMA-HEAP is a DMABUF allocator; however, RK-DMA-HEAP supports both DMABUF allocation and regular page array allocation.

4.1 rk-dma-heap-info

The debug node for RK-DMA-HEAP is `/proc/rk_dma_heap/rk-dma-heap-info`. Reading this node outputs extensive information as follows:

```
# cat /proc/rk_dma_heap/rk-dma-heap-info
dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5656
    Alloc by (vmpi                ) [0x03e1b000-0x03e33fff] 0x00019000 (100 KiB)
        Attached Devices:
Total 0 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5655
    Alloc by (vmpi                ) [0x03e02000-0x03e1afff] 0x00019000 (100 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached
```

```
dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5654
    Alloc by (vmpi                ) [0x03d96000-0x03e01fff] 0x0006c000 (432 KiB)
        Attached Devices:
Total 0 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5653
    Alloc by (vmpi                ) [0x03d86000-0x03d95fff] 0x00010000 (64 KiB)
        Attached Devices:
Total 0 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5652
    Alloc by (vmpi                ) [0x03d76000-0x03d85fff] 0x00010000 (64 KiB)
        Attached Devices:
        ffa50000.rkvenc
        ffa50000.rkvenc
Total 2 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5637
    Alloc by (vmpi                ) [0x03c76000-0x03d75fff] 0x00100000 (1024 KiB)
        Attached Devices:
        ffa50000.rkvenc
        ffa50000.rkvenc
Total 2 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5636
    Alloc by (vmpi                ) [0x03c74000-0x03c75fff] 0x00002000 (8 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5635
    Alloc by (vmpi                ) [0x03c72000-0x03c73fff] 0x00002000 (8 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5626
    Alloc by (vmpi                ) [0x03c40000-0x03c71fff] 0x00032000 (200 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5625
    Alloc by (vmpi                ) [0x03c3f000-0x03c3ffff] 0x00001000 (4 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5624
    Alloc by (vmpi                ) [0x03c37000-0x03c3efff] 0x00008000 (32 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached
```

```
dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5623
    Alloc by (vmpi                ) [0x03c36000-0x03c36fff] 0x00001000 (4 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5622
    Alloc by (vmpi                ) [0x03c2e000-0x03c35fff] 0x00008000 (32 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5621
    Alloc by (vmpi                ) [0x03b64000-0x03c2dfff] 0x000ca000 (808 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5610
    Alloc by (vmpi                ) [0x03acf000-0x03b63fff] 0x00095000 (596 KiB)
        Attached Devices:
        ffa00000.rkisp
        ffa50000.rkvenc
        ffa00000.rkisp
        ffa00000.rkisp
        ffa00000.rkisp
Total 5 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5609
    Alloc by (vmpi                ) [0x03ac0000-0x03acefff] 0x0000f000 (60 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5608
    Alloc by (vmpi                ) [0x03ab1000-0x03abffff] 0x0000f000 (60 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5607
    Alloc by (vmpi                ) [0x038ef000-0x03ab0fff] 0x001c2000 (1800 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5606
    Alloc by (vmpi                ) [0x038ed000-0x038eefff] 0x00002000 (8 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached
```

```
dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5605
    Alloc by (vmpi                ) [0x038b3000-0x038ecfff] 0x0003a000 (232 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5604
    Alloc by (vmpi                ) [0x038b1000-0x038b2fff] 0x00002000 (8 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5603
    Alloc by (vmpi                ) [0x03877000-0x038b0fff] 0x0003a000 (232 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5602
    Alloc by (vmpi                ) [0x0324e000-0x03876fff] 0x00629000 (6308 KiB)
        Attached Devices:
        ffa50000.rkvenc
Total 1 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5586
    Alloc by (ff660000.npu        ) [0x0308c000-0x030f7fff] 0x0006c000 (432 KiB)
        Attached Devices:
Total 0 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5585
    Alloc by (ff660000.npu        ) [0x03087000-0x0308bfff] 0x00005000 (20 KiB)
        Attached Devices:
Total 0 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5584
    Alloc by (ff660000.npu        ) [0x03082000-0x03086fff] 0x00005000 (20 KiB)
        Attached Devices:
Total 0 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5583
    Alloc by (ff660000.npu        ) [0x0307d000-0x03081fff] 0x00005000 (20 KiB)
        Attached Devices:
Total 0 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5582
    Alloc by (ff660000.npu        ) [0x0306b000-0x0307cfff] 0x00012000 (72 KiB)
        Attached Devices:
Total 0 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5581
    Alloc by (ff660000.npu        ) [0x03059000-0x0306afff] 0x00012000 (72 KiB)
        Attached Devices:
Total 0 devices attached
```

```
dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5580
    Alloc by (ff660000.npu        ) [0x03047000-0x03058fff] 0x00012000 (72 KiB)
        Attached Devices:
Total 0 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5579
    Alloc by (ff660000.npu        ) [0x02e5f000-0x03046fff] 0x001e8000 (1952 KiB)
        Attached Devices:
Total 0 devices attached

dma-heap:<rk-dma-heap-cma> -dmabuf  i_ino = 5328
    Alloc by (vmpi                ) [0x02800000-0x02802fff] 0x00003000 (12 KiB)
        Attached Devices:
Total 0 devices attached

dma-heap:<rk-dma-heap-cma> -non dmabuf
    Alloc by (ffa00000.rkisp      ) [0x02803000-0x02813fff] 0x00011000 (68 KiB)
dma-heap:<rk-dma-heap-cma> -non dmabuf
    Alloc by (ffa00000.rkisp      ) [0x02814000-0x02824fff] 0x00011000 (68 KiB)
dma-heap:<rk-dma-heap-cma> -non dmabuf
    Alloc by (ffa00000.rkisp      ) [0x02825000-0x02825fff] 0x00001000 (4 KiB)
dma-heap:<rk-dma-heap-cma> -non dmabuf
    Alloc by (ffa00000.rkisp      ) [0x02826000-0x02826fff] 0x00001000 (4 KiB)
dma-heap:<rk-dma-heap-cma> -non dmabuf
    Alloc by (ffa00000.rkisp      ) [0x02827000-0x02dddfff] 0x005b7000 (5852 KiB)
dma-heap:<rk-dma-heap-cma> -non dmabuf
    Alloc by (ffa00000.rkisp      ) [0x02dde000-0x02e4efff] 0x00071000 (452 KiB)
dma-heap:<rk-dma-heap-cma> -non dmabuf
    Alloc by (rkisp-vir0          ) [0x02e4f000-0x02e52fff] 0x00004000 (16 KiB)
dma-heap:<rk-dma-heap-cma> -non dmabuf
    Alloc by (rkisp-vir0          ) [0x02e53000-0x02e56fff] 0x00004000 (16 KiB)
dma-heap:<rk-dma-heap-cma> -non dmabuf
    Alloc by (rkisp-vir0          ) [0x02e57000-0x02e5afff] 0x00004000 (16 KiB)
dma-heap:<rk-dma-heap-cma> -non dmabuf
    Alloc by (rkisp-vir0          ) [0x02e5b000-0x02e5efff] 0x00004000 (16 KiB)
dma-heap:<rk-dma-heap-cma> -non dmabuf
    Alloc by (ffa00000.rkisp      ) [0x030f8000-0x03249fff] 0x00152000 (1352 KiB)
dma-heap:<rk-dma-heap-cma> -non dmabuf
    Alloc by (ffa00000.rkisp      ) [0x0324a000-0x0324dfff] 0x00004000 (16 KiB)

Total : 0x1634000 (22736 KiB)
```

Generally, use 'Alloc' to filter allocation information.

```
# cat /proc/rk_dma_heap/rk-dma-heap-info |grep Alloc
        Alloc by (vmpi                ) [0x03e1b000-0x03e33fff] 0x00019000 (100 KiB)
        Alloc by (vmpi                ) [0x03e02000-0x03e1afff] 0x00019000 (100 KiB)
        Alloc by (vmpi                ) [0x03d96000-0x03e01fff] 0x0006c000 (432 KiB)
        Alloc by (vmpi                ) [0x03d86000-0x03d95fff] 0x00010000 (64 KiB)
        Alloc by (vmpi                ) [0x03d76000-0x03d85fff] 0x00010000 (64 KiB)
        Alloc by (vmpi                ) [0x03c76000-0x03d75fff] 0x00100000 (1024 KiB)
        Alloc by (vmpi                ) [0x03c74000-0x03c75fff] 0x00002000 (8 KiB)
```

```
        Alloc by (vmpi                ) [0x03c72000-0x03c73fff] 0x00002000 (8 KiB)
        Alloc by (vmpi                ) [0x03c40000-0x03c71fff] 0x00032000 (200 KiB)
        Alloc by (vmpi                ) [0x03c3f000-0x03c3ffff] 0x00001000 (4 KiB)
        Alloc by (vmpi                ) [0x03c37000-0x03c3efff] 0x00008000 (32 KiB)
        Alloc by (vmpi                ) [0x03c36000-0x03c36fff] 0x00001000 (4 KiB)
        Alloc by (vmpi                ) [0x03c2e000-0x03c35fff] 0x00008000 (32 KiB)
        Alloc by (vmpi                ) [0x03b64000-0x03c2dfff] 0x000ca000 (808 KiB)
        Alloc by (vmpi                ) [0x03acf000-0x03b63fff] 0x00095000 (596 KiB)
        Alloc by (vmpi                ) [0x03ac0000-0x03acefff] 0x0000f000 (60 KiB)
        Alloc by (vmpi                ) [0x03ab1000-0x03abffff] 0x0000f000 (60 KiB)
        Alloc by (vmpi                ) [0x038ef000-0x03ab0fff] 0x001c2000 (1800 KiB)
        Alloc by (vmpi                ) [0x038ed000-0x038eefff] 0x00002000 (8 KiB)
        Alloc by (vmpi                ) [0x038b3000-0x038ecfff] 0x0003a000 (232 KiB)
        Alloc by (vmpi                ) [0x038b1000-0x038b2fff] 0x00002000 (8 KiB)
        Alloc by (vmpi                ) [0x03877000-0x038b0fff] 0x0003a000 (232 KiB)
        Alloc by (vmpi                ) [0x0324e000-0x03876fff] 0x00629000 (6308 KiB)
        Alloc by (ff660000.npu        ) [0x0308c000-0x030f7fff] 0x0006c000 (432 KiB)
        Alloc by (ff660000.npu        ) [0x03087000-0x0308bfff] 0x00005000 (20 KiB)
        Alloc by (ff660000.npu        ) [0x03082000-0x03086fff] 0x00005000 (20 KiB)
        Alloc by (ff660000.npu        ) [0x0307d000-0x03081fff] 0x00005000 (20 KiB)
        Alloc by (ff660000.npu        ) [0x0306b000-0x0307cfff] 0x00012000 (72 KiB)
        Alloc by (ff660000.npu        ) [0x03059000-0x0306afff] 0x00012000 (72 KiB)
        Alloc by (ff660000.npu        ) [0x03047000-0x03058fff] 0x00012000 (72 KiB)
        Alloc by (ff660000.npu        ) [0x02e5f000-0x03046fff] 0x001e8000 (1952 KiB)
        Alloc by (vmpi                ) [0x02800000-0x02802fff] 0x00003000 (12 KiB)
        Alloc by (ffa00000.rkisp      ) [0x02803000-0x02813fff] 0x00011000 (68 KiB)
        Alloc by (ffa00000.rkisp      ) [0x02814000-0x02824fff] 0x00011000 (68 KiB)
        Alloc by (ffa00000.rkisp      ) [0x02825000-0x02825fff] 0x00001000 (4 KiB)
        Alloc by (ffa00000.rkisp      ) [0x02826000-0x02826fff] 0x00001000 (4 KiB)
        Alloc by (ffa00000.rkisp      ) [0x02827000-0x02dddfff] 0x005b7000 (5852 KiB)
        Alloc by (ffa00000.rkisp      ) [0x02dde000-0x02e4efff] 0x00071000 (452 KiB)
        Alloc by (rkisp-vir0          ) [0x02e4f000-0x02e52fff] 0x00004000 (16 KiB)
        Alloc by (rkisp-vir0          ) [0x02e53000-0x02e56fff] 0x00004000 (16 KiB)
        Alloc by (rkisp-vir0          ) [0x02e57000-0x02e5afff] 0x00004000 (16 KiB)
        Alloc by (rkisp-vir0          ) [0x02e5b000-0x02e5efff] 0x00004000 (16 KiB)
        Alloc by (ffa00000.rkisp      ) [0x030f8000-0x03249fff] 0x00152000 (1352 KiB)
        Alloc by (ffa00000.rkisp      ) [0x0324a000-0x0324dfff] 0x00004000 (16 KiB)
```

Format: Alloc by ([dmabuf name]) [0xaaaaaaaa-0xbbbbbbbb] 0xzzzzzzzz (dd KiB)

1. [dmabuf name]: The thread name that allocates memory in the Driver is set to the dmabuf name by default; this default value is overridden if a specific string is specified by each module. For example, vmpi is the thread name in rockit that calls rk-dma-heap.
2. [0xaaaaaaaa-0xbbbbbbbb]: 0xaaaaaaaa is the start address (physical address) of the memory region, and 0xbbbbbbbb is the end address (physical address).
3. 0xzzzzzzzz: The hexadecimal value of the memory region size.
4. dd KiB: The decimal value of the memory region size converted to KiB.